SEMESTER PROJECT PROPOSAL

CampusFinder

Campus Lost and Found System with Image Matching

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Document	Project Proposal (Business Case Charter)

1. Background and Introduction

Every university campus loses items every day: ID cards, calculators, water bottles, earphones, wallets, USB drives, and keys. Right now, most of these items end up with the security desk or are announced in student WhatsApp groups, where the message is buried within minutes. The owner rarely finds the item, and the finder has no reliable way to return it.

CampusFinder is a web-based Lost and Found system for a university campus. A student uploads a photo and short description of the lost item, security staff upload photos of items that were found, and the system automatically compares the images and shows the most likely matches. The owner then submits a claim, and the staff verifies it before handing over the item.

2. Problem Statement

Campus lost and found is handled manually through notice boards, a security register, and group chats. This causes the following problems:

- **No matching:** Found items and lost reports are never matched with each other in a structured way.
- **Vague descriptions:** Text descriptions like "black bottle" or "blue pouch" are too vague, and many items look the same.
- **No claim verification:** Anyone can walk up and claim an item because there is no proof-of-ownership step.
- **No tracking:** There is no record of how many items were lost, returned, or left unclaimed.

The project solves this by providing image-based matching together with a controlled claim and verification workflow.

3. Business Objectives

- Reduce the average time to locate a lost item from several days to under 24 hours.
- Return at least 60% of reported items to their owners within one month of deployment.
- Show the top 5 visually similar found items for every lost-item report in under 10 seconds.
- Prevent wrong handovers by requiring staff-approved claim verification for every item.
- Give administration a dashboard with lost, found, returned, and unclaimed item statistics.

4. Preliminary Requirements

4.1 Functional Requirements

ID	Requirement
FR-01	A student shall be able to register and log in using a university email address.
FR-02	A student shall be able to submit a lost-item report with 1 to 3 photos, category, location, and date.
FR-03	Security staff shall be able to submit a found-item entry with 1 to 3 photos, category, and storage location.

ID	Requirement
FR-04	The system shall compute a similarity score between each lost report and all open found items.
FR-05	The system shall display the top 5 matches for a lost report, ranked by similarity score.
FR-06	The system shall notify the student by email and in-app alert when a match above the set threshold is found.
FR-07	A student shall be able to submit a claim by answering at least 2 ownership questions about the item.
FR-08	Security staff shall approve or reject a claim and record the handover date.
FR-09	The administrator shall be able to view reports of lost, found, returned, and unclaimed items by month.
FR-10	The administrator shall be able to manage user accounts and the match threshold.

4.2 Non-Functional Requirements

ID	Requirement
NFR-01	Performance: Matching results shall be displayed within 10 seconds for a database of up to 5,000 found items.
NFR-02	Accuracy: The correct item shall appear in the top 5 results in at least 75% of test cases.
NFR-03	Security: Passwords shall be stored hashed, and only staff accounts shall access claim approval.
NFR-04	Privacy: Contact details of the finder or owner shall never be shown to other students.
NFR-05	Usability: A new student shall be able to submit a lost report within 3 minutes without training.
NFR-06	Availability: The system shall be available at least 95% of the time during semester days.

5. Stakeholders

Stakeholder	Needs and Concerns
Students	Quickly report lost items, see matches, and claim items easily. Concerned about privacy of personal details.
Security Staff	Easy upload of found items, clear claim verification, and a record of every handover.
Administration	Statistics, accountability, and control over users and settings.
Faculty and Employees	Same reporting and claim access as students with staff-level email verification.

6. Current Situation and Opportunity

At present, a student who loses an item has to visit the security desk in person, scroll through group chats, or ask friends. There is no search, no photo comparison, and no history. Many universities already have the same problem, but most ready-made solutions are general apps that do not fit a campus workflow with security staff verification.

The opportunity is to use image similarity techniques, which are now practical with pre-trained deep learning models, to match items automatically and combine this with a verification workflow designed for a campus.

Benchmarking

The table below compares the existing ways of handling lost items with the proposed system.

Approach	Photo Upload	Auto Image Matching	Claim Verification	Item Tracking	Admin Reports
Notice board / security register	No	No	No	Limited	No
WhatsApp / social media groups	Yes	No	No	No	No
Generic lost-and-found apps	Yes	Varies	Limited	Yes	Limited
CampusFinder (proposed)	Yes	Yes	Yes	Yes	Yes

Note: The comparison is based on the general categories of existing approaches, not on specific products.

7. Complex Computing Problem (CCP) Justification

This project is not just a database with add and search functions. The following CCP characteristics genuinely apply:

CCP Characteristic	Project-Specific Evidence
Depth of Knowledge	Matching needs image feature extraction using a pre-trained convolutional neural network and similarity measurement such as cosine similarity. This goes beyond routine database coursework.
Conflicting Requirements	A low threshold finds more matches but shows wrong items; a high threshold misses real ones. Also, quick claims conflict with strict ownership verification and privacy of users.
Depth of Analysis	There is no single obvious solution. Different feature extractors and thresholds must be compared on test images and selected using top-5 accuracy.
Extent of Stakeholder Involvement	Students, security staff, administration, and faculty have different permissions, workflows, and privacy expectations.
Interdependence	Image upload, preprocessing, feature extraction, matching engine, notifications, claim workflow, and admin dashboard depend on each other. A fault in one affects all.
Consequences	A wrong match or weak verification can hand a person's belongings (ID card, wallet, laptop) to the wrong individual and lead to loss, disputes, or privacy leaks.

8. Options and Recommendations

Option	Description	Evaluation
A: Manual register	Continue the paper register and notice board.	Cheap but slow, no matching, no tracking.

Option	Description	Evaluation
B: Text-only portal	Web portal where users search items by text description.	Easy to build, but vague descriptions give poor results.
C: Image-matching portal (recommended)	Web portal with photo upload, automatic similarity matching, claim verification, and dashboard.	Best match quality and accountability; needs more analysis and testing effort.

Recommendation: Option C is recommended because it solves the main problem of vague descriptions and also provides the verification and tracking that the other options lack.

9. Schedule Estimate

Weeks	Activity	Deliverable
1 - 2	Requirements gathering, stakeholder analysis, and proposal	Proposal and SRS draft
3 - 4	System design: architecture, database, and UML diagrams	Design document
5 - 7	Core modules: authentication, lost and found submission	Working basic portal
8 - 10	Image feature extraction and matching engine	Matching module
11 - 12	Claim verification, notifications, and admin dashboard	Complete workflow
13 - 14	Testing with sample images, threshold tuning, and bug fixing	Test report
15 - 16	Final documentation, demo, and presentation	Final submission

10. Potential Risks and Mitigation Plan

Risk	Impact	Mitigation
Poor image quality (dark, blurred photos)	Wrong or missed matches	Guide users to take clear photos; allow up to 3 photos; add basic image preprocessing.
Similar-looking items (e.g., identical bottles)	High false-match rate	Combine image score with category, location, and date filters.
False claims by other people	Item handed to wrong person	Ownership questions and mandatory staff approval.
Privacy leak of user data	Loss of user trust	Hide contact details, hash passwords, role-based access.
Not enough test images	Weak accuracy measurement	Build own sample set of 100+ item photos.
Time shortage	Incomplete features	Prioritise core matching and claim workflow first; keep dashboard simple.

11. Conclusion

CampusFinder addresses a real and everyday campus problem with a design that needs image analysis, careful trade-offs, and several interacting components. It satisfies multiple CCP characteristics and gives a measurable outcome: the correct item appearing among the top 5 matches. I request approval to proceed with this project.